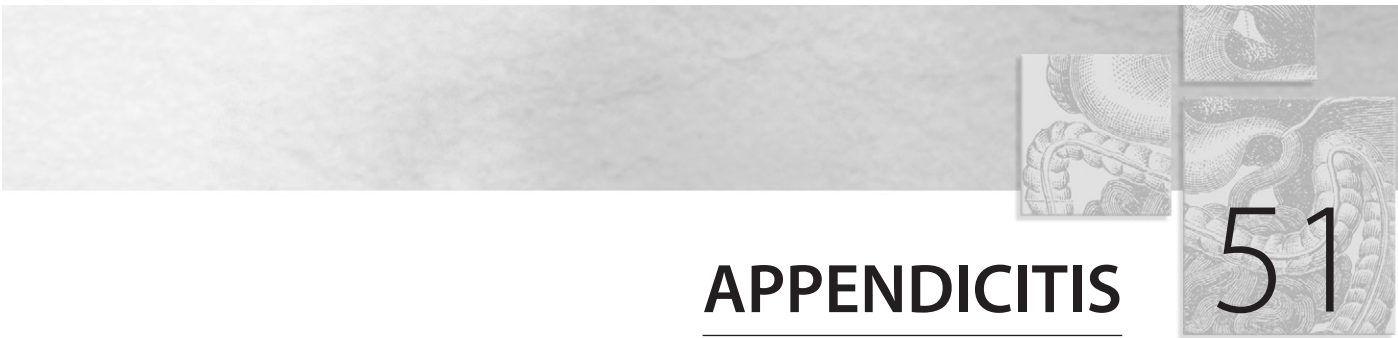




APPENDICITIS

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Acute appendicitis is one of the most common surgical conditions afflicting children and adults. Approximately 250,000 cases occur annually in the United States, with the highest incidence in patients 10 to 19 years old. The lifetime risk of developing appendicitis has been estimated at 8.6% for males and 6.7% for females, with a lifetime risk of undergoing an appendectomy estimated at 12% for males and 23% for females.¹ Appendiceal perforation in the pediatric population is more common than in adults, occurring in up to 50% of patients. In children less than 5 years of age, the perforation rate is 65%, and in those less than 2 years of age, the rate is 95%.²

A thorough understanding of this common surgical condition is important for all health care providers, who should be able to recognize and promptly refer patients with possible appendicitis for further evaluation to avoid complications related to a delay in diagnosis.

EMBRYOLOGY AND ANATOMY

The development of the appendix begins during the eighth week of gestation. The appendix, which initially projects from the apex of the cecum, is progressively displaced medially toward the ileocecal valve.³ Lymph nodes begin to develop by the seventh month of gestation and continue to proliferate until puberty, after which they start to regress.⁴ In the adult, the appendix varies in length, but averages about 8 cm; it is relatively longer and narrower in children. It has its own mesentery, called the mesoappendix, which contains the appendicular artery, a branch of the ileocolic artery, which arises from the superior mesenteric artery (Figure 51-1). The ileocolic vein drains blood from the cecum and appendix and enters the superior mesenteric vein to drain into the portal vein.

PATHOPHYSIOLOGY

The etiology of appendicitis was initially outlined in 1939 by Wangenstein and Dennis in their paper “Experimental Proof of the Obstructive Origin of Appendicitis in Man.”⁶ Appendicitis is initiated by obstruction of the appendiceal lumen. When the lumen of the appendix becomes obstructed, the flow of normal mucosal secretions is inhibited. This leads to increased intraluminal pressure and compromised venous drainage, which leads to ischemic breakdown of the mucosa. Simultaneously, luminal bacteria, typically *Escherichia coli*, *Bacteroides fragilis*, *Pseudomonas aeruginosa*, and *Clostridium* species, proliferate and traverse the appendiceal wall. This sequence of events results in an acute infection, gangrene, and ultimately perforation.

As the inflammatory process progresses, the omentum as well as adjacent loops of bowel may surround the inflamed area. If the disease progresses to perforation, these surrounding structures act as barriers that wall off the infection and

frequently prevent free perforation and instead result in the formation of an inflammatory mass (phlegmon) or abscess. The formation of a phlegmon or abscess is central to the success of nonoperative management of perforated appendicitis, which is discussed later. On the other hand, free perforation with diffuse peritonitis occurs when the appendix perforates before it is sufficiently isolated by surrounding tissue.

HISTOLOGY

The appendix is composed of the same four layers as the rest of the intestine: the mucosa, submucosa, muscularis propria, and serosa. The mucosa is made of a surface layer of epithelial cells, a loose connective tissue layer known as the lamina propria, as well as the muscularis mucosa that separates the mucosa from the submucosa. The epithelial surface contains a combination of columnar cells with basally located nuclei, mucus-secreting goblet cells, and absorptive cells. In addition, the epithelial surface contains Paneth cells, which harbor lysozyme-containing granules and are thought to contribute to the intestinal defense system against bacteria, as well as neuroendocrine cells. These neuroendocrine cells are the origin of the carcinoid tumor, which is the most common appendiceal neoplasm. The lamina propria, just underneath the surface epithelial layer, contains the crypts of Lieberkühn as well as lymphoid follicles with germinal centers. The next layer is the submucosa, which contains a rich network of blood vessels, lymphatics, and nerves. The network of ganglion cells and Schwann cells found within this layer is known as Meissner's plexus. The next layer is the muscularis propria, which consists of two layers of smooth muscle. The muscle fibers of the inner layer are arranged in a circular fashion, whereas the fibers of the outer layer are arranged longitudinally. Another neural network, known as Auerbach's plexus, lies between these muscle layers. The outermost layer of the appendiceal wall, the serosa, is made up of a band of fibrous tissue with an overlying layer of cuboidal mesothelial cells.

CLINICAL PRESENTATION

Early diagnosis is vital to improving outcome and avoiding the morbidity associated with appendiceal perforation. Although laboratory and radiologic studies are useful aids in making the diagnosis, the key components are the history and physical examination.

The “typical” presentation of periumbilical pain that migrates to the right lower quadrant is a reflection of the progression of inflammation and pain from visceral to somatic pathways. Initially the contraction of the appendix against an obstructed lumen results in activation of visceral afferent fibers that enter the spine at the level of T10, causing a vague pain that is referred to the periumbilical area. As the inflammation becomes